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Resolution Choices in Rock Classification: A Comparative Study of Core Logging and XRF Techniques

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Abstract

Unconventional reservoirs including shale gas, tight oil, and organic-rich mudstones have emerged as key contributors to global energy supply over the last two decades, owing to advances in horizontal drilling and hydraulic fracturing. However, the inherent complexity and fine-scale heterogeneity of these formations pose significant challenges for accurate reservoir characterization. Sampling resolution is a critical but often underappreciated factor in both core analysis and X-ray fluorescence (XRF) geochemical characterization. In this study, we systematically evaluate the effects of sampling interval by comparing core logger and XRF data at 1, 3, 5, 15, and 30 cm intervals. We analyze the impacts on facies classification, continuous property measurements, and the detection of organic-rich (TOC) zones. Our results show that, although mean property values and bulk facies proportions are relatively stable across different resolutions, local agreement in facies assignment and the ability to detect thin, TOC-rich beds decline sharply at coarser intervals. High-resolution scanning (≤ 3 cm) proves essential for reliable advanced interpretation, including sweet spot identification, geomechanical assessment, and completion strategy. Our findings provide both quantitative benchmarks and practical guidelines for future core characterization in resource plays, emphasizing the need for careful selection of sampling resolution to avoid misinterpretation and missed opportunities.

Key words: Rock Classification, High-Resolution Analysis, Multi-Sensor Core Logging, X-ray Fluorescence, Resolution Optimization

Introduction

Unconventional reservoirs, such as shale gas, tight oil, and organic-rich mudstones, have fundamentally transformed the energy landscape, enabling access to vast hydrocarbon resources previously deemed uneconomical. Unlike conventional reservoirs, unconventional formations are characterized by extremely low matrix permeability and a high degree of mineralogical and geochemical heterogeneity, often at the scale of millimeters to centimeters (e.g., [Macquaker et al., 2007](#); [Suarez-Rivera et al., 2012](#); [Pasqualetti, M.J., 2012](#); [Lazar et al., 2015](#)). The spatial variability of lithofacies, porosity, permeability, and organic

matter content directly influences not only reservoir quality but also hydraulic fracturing effectiveness, well performance, and resource estimation.

To address these complexities, advanced core analysis and X-ray fluorescence (XRF) geochemistry have become indispensable tools for subsurface evaluation (Rowe et al., 2012; Chan et al., 2022; Chan et al., 2023; Albensaad et al., 2024; Chan et al., 2025). High-resolution core scanning collecting physical and chemical measurements at centimeter or even millimeter intervals enable the detection of subtle bedding, mineralogical gradients, and organic-rich "sweet spots". However, logistical, time, and cost constraints often result in the acquisition of core and XRF data at much coarser intervals (e.g., 15 or 30 cm), especially in older or large-scale datasets). The quantitative impact of such down sampling on reservoir characterization is not fully understood, and there is limited guidance in the literature on the minimum resolution required for robust interpretation.

Recent studies have raised concerns that coarse sampling intervals may obscure thin beds, mask sharp geochemical transitions, and introduce biases in facies classification and resource estimation. For unconventional reservoirs, where subtle heterogeneity can control fluid flow, fracture propagation, and sweet spot development, the implications of insufficient resolution can be significant. Advanced data analytics, machine learning, and geostatistical techniques can only partially mitigate these issues if the underlying data lacks the necessary detail.

In this study, we provide a comprehensive assessment of how sampling resolution influences the outcome of core logging and XRF-based geochemical analysis in an unconventional reservoir. By systematically comparing datasets collected or downsampled to 1, 3, 5, 15, and 30 cm intervals, we aim to: (1) quantify the loss of detail and accuracy in facies classification, (2) evaluate the effect on continuous property measurements and their geochemical "fingerprints," and (3) determine the sensitivity of TOC-rich interval detection to interval spacing. Our goal is to deliver practical recommendations for core analysis workflows in unconventional plays, supported by robust quantitative evidence and grounded in published best practices.

Materials and Method

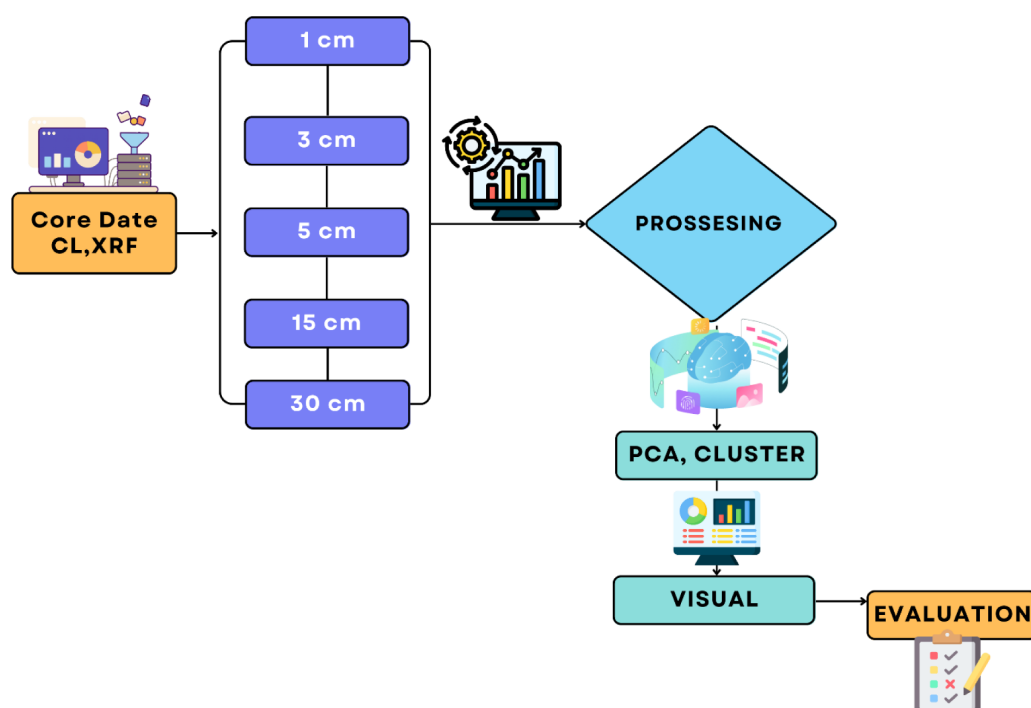


Figure 1—Methodology workflow.

Core and XRF Data Acquisition

Core and XRF data were obtained from a representative unconventional reservoir section, targeting intervals with significant lithological and geochemical variability. The core was described visually and scanned using a multi-sensor core logger to obtain high-resolution physical property measurements, including density, magnetic susceptibility, and natural gamma. Geochemical analysis was performed using a Handheld XRF analyzer providing major and trace element concentrations at high spatial resolution. The raw high-resolution dataset (typically at 1 cm spacing) was then systematically downsampled by averaging or resampling to generate equivalent datasets at 3, 5, 15, and 30 cm intervals.

Sample preparation followed standard protocols to minimize contamination and maximize measurement reliability. All instruments were Calibrated using certified reference materials, and analytical precision was monitored through replicate measurements and quality control checks.

Data Processing and Analysis

All geochemical and physical property data collected from core samples were subjected to a comprehensive data processing and analysis workflow to ensure quality, consistency, and suitability for advanced interpretation.

Data Cleaning and Quality Control. Raw data from XRF measurements and physical property logs (density, P-wave velocity) were first inspected for inconsistencies, outliers, and missing values. Obvious data entry errors and non-physical measurements were identified and removed. Quality control procedures included:

- Cross-validation with reference materials and duplicate measurements.
- Visual inspection of log curves to identify and correct spikes or drift.
- Interpolation to address small gaps, with missing values retained for later imputation if needed. (Chan et al., 2025).

Depth Alignment and Resolution Adjustment. To enable direct comparison across datasets and resolutions, all measurements were depth-matched using a standardized core depth reference. Data were resampled to target intervals of 1 cm, 3 cm, 5 cm, 15 cm, and 30 cm using averaging or interpolation methods, depending on the original data density and the nature of the variable. This allowed for systematic evaluation of the effects of sampling resolution on geological interpretation.

Normalization and Transformation. Elemental concentrations and physical property values were normalized to account for differences in measurement scales and to facilitate statistical analyses. This included Z-score normalization for machine learning workflows and, where appropriate, log-transformation of highly skewed variables to reduce the influence of outliers.

Statistical and Machine Learning Analyses. A suite of statistical and machine learning techniques was applied to evaluate the impact of sampling resolution on data structure and lithofacies classification:

- Descriptive statistics and boxplots were generated for each variable and resolution to visualize changes in distribution, variance, and outlier occurrence.
- Principal Component Analysis (PCA) was performed to explore patterns of covariance among variables and assess how resolution affects multivariate structure.
- Clustering algorithms (such as k-means) were applied to group samples based on geochemical and physical similarity, enabling quantitative facies classification.
- Lithofacies classification outputs at each resolution were compared to assessing reproducibility, detail, and consistency of geological interpretation

Software. All data processing, statistical analysis, and visualization were conducted using Microsoft Excel, and Python (with packages including pandas, numpy, scikit-learn, and matplotlib), supplemented by Data management and initial QC were performed in Microsoft Excel where necessary.

Results and Discussion

Data Resolution and Sampling Density

At the finest sampling interval (1 cm), a total of 5,674 data points were collected. As the resolution becomes coarser, the number of data points drops sharply:

- 3 cm: 1,892 data points (~33% of 1 cm)
- 5 cm: 1,135 data points (~20% of 1 cm)
- 15 cm: 379 data points (~7% of 1 cm)
- 30 cm: 190 data points (~3% of 1 cm)

This dramatic reduction highlights how higher resolution sampling substantially increases the granularity and completeness of the dataset.

For rock classification purposes, fine-scale data are crucial for detecting subtle lithological changes, thin beds, or mineralogical transitions that may be completely missed at coarser intervals.

Moreover, the importance of high-resolution data becomes even more pronounced in advanced data analytics and machine learning workflows. Techniques such as Principal Component Analysis (PCA) and clustering rely heavily on abundant and detailed datasets to uncover hidden patterns, distinguish between subtle facies, and improve classification accuracy. Coarse sampling not only reduces the statistical power of these methods but can also obscure key geological features, leading to less reliable interpretations.

In summary, while fine resolution increases the time and resources needed for data collection and processing, it provides clear benefits for both traditional geological analysis and modern machine learning approaches enabling more accurate, detailed, and actionable rock classification.

Comparison of Lithofacies Classification Across Sampling Resolutions

The impact of sampling resolution on lithofacies classification and the statistical properties of key geochemical and physical variables is illustrated in Figures 2 and 3.

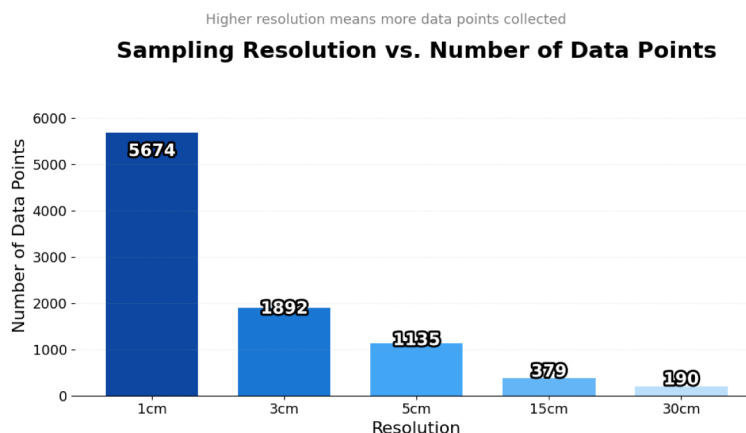


Figure 2—Data point count at each resolution. Finer sampling (e.g., 1 cm) produces much higher data density, emphasizing the effect of resolution on dataset size.

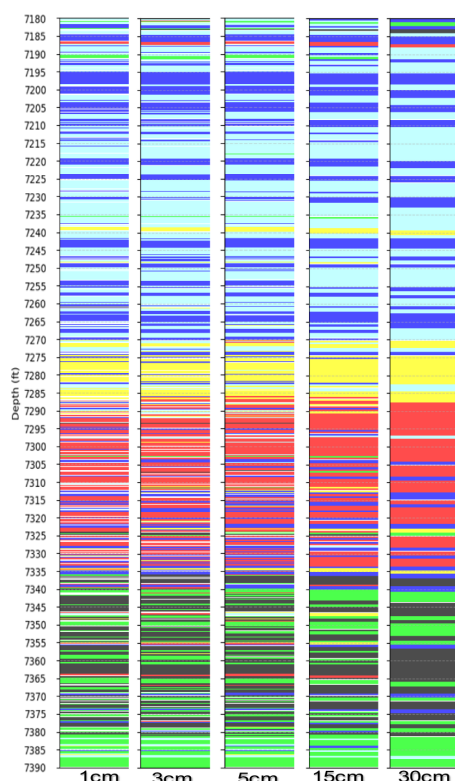


Figure 3—Lithofacies classification at different sampling resolutions (1–30 cm) over the same depth interval (7180–7390 ft). Coarser resolutions yield thicker units and less fine-scale detail.

Figure 3 presents lithofacies classification outputs for the same depth interval (7180–7390 ft) using multiple sampling resolutions (1 cm, 3 cm, 5 cm, 15 cm, and 30 cm, from left to right). Each colored band indicates a distinct lithofacies. At the highest resolution (1 cm), the classification reveals fine stratigraphic detail, including thin beds and subtle facies transitions. As the sampling interval increases, many of these fine-scale features become undetectable, with coarser resolutions (especially 15 and 30 cm) only resolving the thickest and most prominent units. This clearly demonstrates how lower resolutions can oversimplify the geological picture by smoothing over critical small-scale heterogeneities.

To further quantify these effects, Figure 4 shows boxplots for major geochemical and physical properties (such as Al, Ca, Si, Ti, Sr, Mo, Ni, Cu, V, P-wave velocity, and density) across the same set of sampling resolutions. These boxplots highlight several important trends:

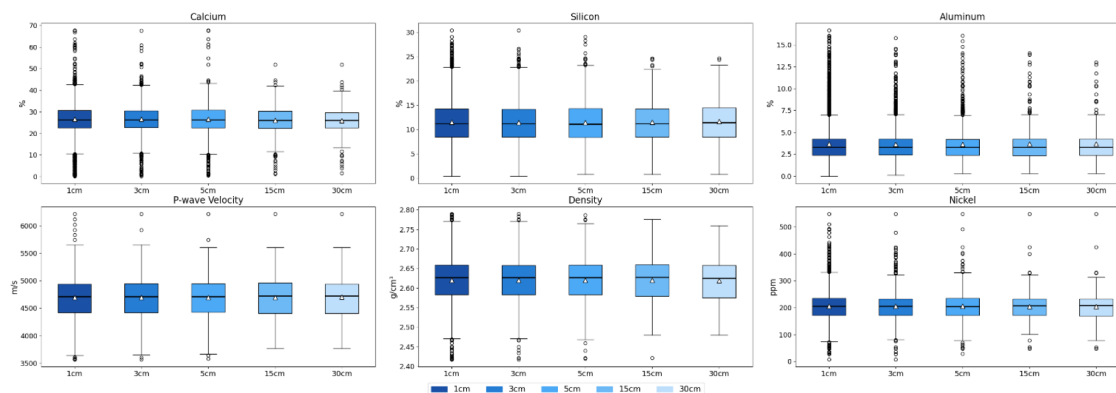


Figure 4—Boxplots showing the distribution of major geochemical and physical variables at five different sampling resolutions (1 cm, 3 cm, 5 cm, 15 cm, and 30 cm). The plots illustrate that as resolution decreases, the data distribution becomes smoother

At fine resolutions (1–3 cm), data distributions retain their full range, capturing numerous outliers and reflecting the real spatial heterogeneity within the formation. This level of detail is crucial for accurately characterizing complex geological features. As the resolution decreases, however, the distributions become narrower, and outliers diminish particularly for elements that exhibit strong local variability. While median values are generally preserved, allowing overall trends to remain visible, much of the detailed variation is lost. Some properties, such as P-wave velocity or bulk density, are relatively stable across different resolutions and show minimal sensitivity to sampling scale. In contrast, trace elements and other highly variable indicators experience a significant reduction in observed variability at coarser resolutions, limiting their interpretive value.

Together, these results underscore the critical role of high-resolution sampling in both lithofacies interpretation and quantitative analysis. Fine-scale data are essential for capturing geological complexity, detecting subtle transitions, and maximizing the effectiveness of machine learning techniques such as PCA and clustering. Lower resolutions, while practical for reducing data volume and processing time, may mask important geological signals and compromise classification accuracy.

Overlay Comparison of Geochemical and Physical Logs at Different Resolutions

Both density and P-wave velocity logs demonstrate that measurements at 1 cm and 3 cm resolutions are virtually indistinguishable throughout the studied interval, faithfully tracking abrupt changes and narrow spikes associated with thin beds or discrete lithological contacts. The 15 cm curve, by contrast, appears somewhat smoother, with subtle features partially muted. In the case of calcium, strontium, and silicon, high-frequency alternations captured at 1 cm and 3 cm intervals often reflect subtle changes between carbonate-rich and siliciclastic or mixed beds. Notably, the Sr and Si logs reveal rapid vertical shifts and thin anomalous zones at high resolution, but at 15 cm sampling, such detail is largely smoothed out short-lived anomalies are merged into background trends, and sharp compositional breaks are transformed into less distinct transitions.

Similarly, aluminum and vanadium sensitive indicators of clay-rich layers and redox conditions, respectively exhibit recurrent, sharp excursions at high resolution, signaling thin beds or laminae with unique mineralogical or geochemical signatures. These features are often unresolved at 15 cm intervals, particularly where multiple peaks occur within a single sampling window, potentially leading to missed identification of thin baffles, flow barriers, or zones of mineralization. Copper, another trace metal, often serves as a marker of diagenetic alteration or paleoenvironmental shifts (Shah, N.S. and Soylak, M., 2022.). At 1 cm and 3 cm resolution, rapid local peaks and troughs are evident, whereas the 15 cm curve appears almost featureless, further illustrating the loss of detail with coarser sampling.

This loss of fine-scale variability at lower resolutions also has a direct impact on advanced data analytics and machine learning workflows. For techniques such as principal component analysis (PCA), clustering, and predictive modeling, detailed input data are critical for distinguishing subtle facies and extracting robust geological patterns. Coarser data reduces the sensitivity and accuracy of these methods, potentially diminishing the reliability of rock typing, reservoir quality prediction, and automated facies classification.

Overall, these findings underscore that high-resolution (1–3 cm) measurements are critical for capturing the full spectrum of subsurface variability. Thin beds less than 10 cm thick which may represent key stratigraphic, reservoir, or geomechanical units are only visible at the finest sampling intervals. Conversely, conventional openhole logs, even at 15 cm sampling (often considered "high-resolution" in industry practice), routinely miss subtle but geologically significant features. This can result in oversimplified facies models, overlooked pay zones, unrecognized barriers to flow, or incomplete geomechanical risk assessments.

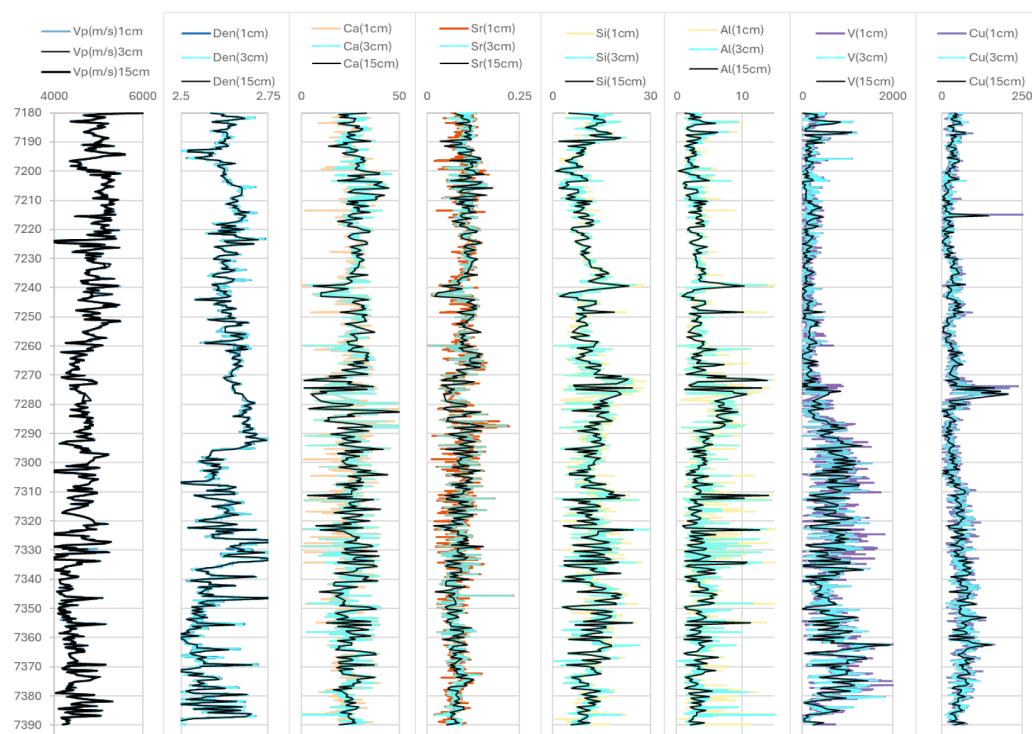


Figure 5—Overlay of logs at all resolutions (1–30 cm) across the depth interval. Higher resolutions capture more stratigraphic detail; lower resolutions show increased smoothing and loss of subtle features.

Agreement Analysis: Resolution Effect on Classification Accuracy

To evaluate how sampling resolution affects the accuracy of lithofacies classification, both local agreement plots and quantitative comparison metrics were employed.

Figure 6 provides a visual summary of where coarser-resolution classifications (3, 5, 15, and 30 cm) agree (green) or disagree (red) with the high-resolution 1 cm reference. This highlights how frequently fine-scale features are captured or missed at each interval.

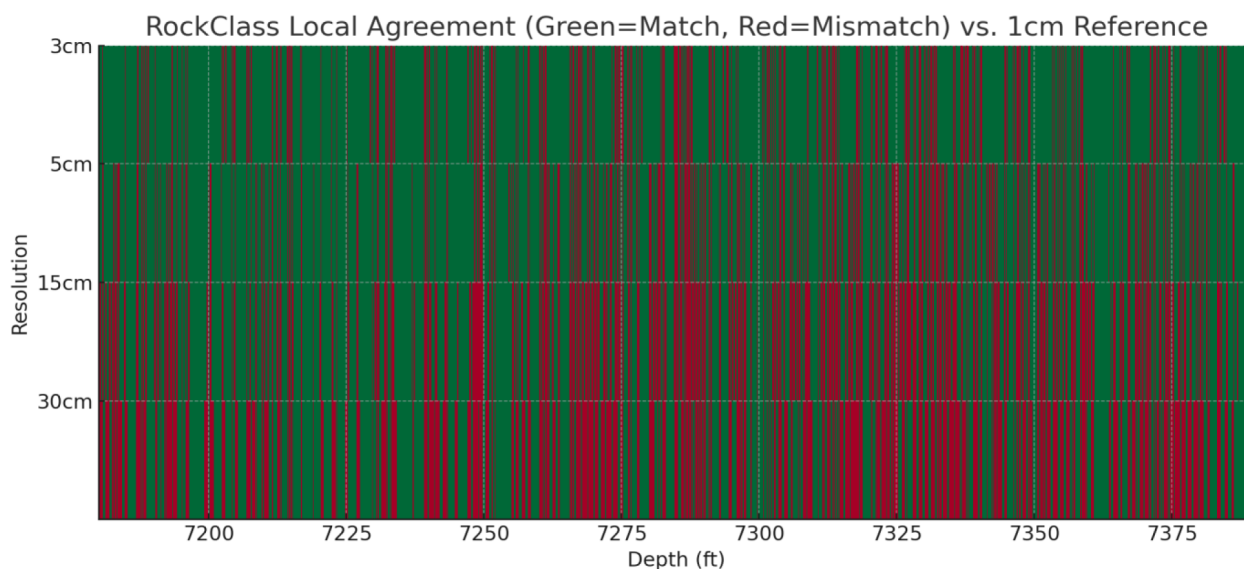


Figure 6—Local agreement between coarser resolutions and the 1 cm reference. Green = match, red = mismatch. Agreement drops notably with an increase in sampling interval.

Table 1 presents the corresponding quantitative results, listing the total number and percentage of matches and mismatches for each resolution, as compared to the 1 cm "ground truth." At 3 cm resolution, there is a high degree of agreement with the 1 cm reference (80%), indicating that most geological features are still accurately captured.

- 5 cm sampling yields good, but slightly reduced agreement (73.9%), with more fine features lost.
- 15 cm and 30 cm resolutions show a sharp drop in agreement (59.7% and 52.6%, respectively), with nearly half of intervals at 30 cm mismatching the high-resolution classification.
- The number of matching intervals decreases steadily as sampling becomes coarser, highlighting the cumulative impact on classification accuracy.

Table 1—Quantitative summary of lithofacies classification agreement and disagreement at different sampling resolutions relative to the 1 cm reference. High-resolution sampling provides much greater classification accuracy

<i>Resolution</i>	<i>% Agreement with 1cm</i>	<i>%Disagreement with 1cm</i>	<i>match</i>	<i>mismatch</i>	<i>Total</i>
3 cm	80%	20%	4630	1044	5674
5 cm	73.90%	26.10%	4363	1311	5674
15 cm	59.70%	40.30%	3501	2173	5674
30 cm	52.60%	47.40%	3098	2576	5674

These results quantitatively confirm the visual trends from Figure 6. While 3 cm and 5 cm resolutions are generally sufficient for capturing major facies boundaries, they increasingly miss thin beds and subtle transitions. Coarse resolutions (15 cm and 30 cm) are inadequate for detailed facies analysis, underscoring the necessity of high-resolution data for robust rock classification and reliable subsurface interpretation. illustrate these trends: high-resolution logs capture thin beds and rapid facies transitions that are progressively merged or missed as intervals become coarser (Kennedy, M., 2015.). The risk of misclassification is particularly acute in intervals with rapid lithological or geochemical change.

Case Study: Enhanced Geological Resolution in the Interval 7335–7385 ft

To further illustrate the effect of sampling resolution on geological interpretation, Figure 7 presents a side-by-side comparison of lithofacies classification for the interval 7335–7390 ft at resolutions of 1 cm, 3 cm, 5 cm, 15 cm, and 30 cm.

At 1 cm and 3 cm resolution, the classification output captures numerous thin beds, sharp lithological contacts, and rapid facies changes, revealing the true complexity of the subsurface. This is especially apparent in several key zones such as around 7338–7342 ft, 7346–7357 ft, and 7360–7367 ft, where thin beds and alternating facies are consistently detected at high resolution.

However, as the resolution coarsens to 5 cm and 15 cm, much of this complexity is lost. In these highlighted intervals, thin layers are merged, and rapid facies transitions become indistinct or are omitted entirely. For example, the complex facies alternations between 7346 and 7357 ft that are clearly resolved at 1 cm and 3 cm are largely smoothed over or missed at coarser intervals. Similarly, in the interval from 7360 to 7367 ft, multiple thin beds apparent at high resolution disappear almost entirely in the 15 cm and 30 cm columns.

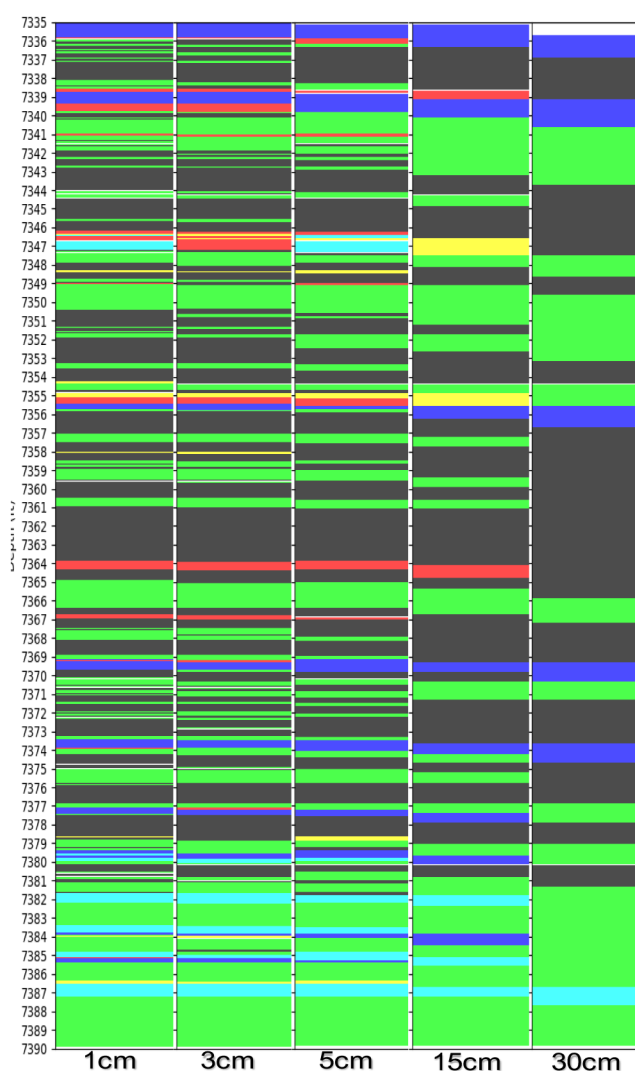


Figure 7—Lithofacies classification from 7335–7390 ft at varying resolutions. Thin beds and sharp boundaries seen at 1–3 cm are lost at coarser intervals, oversimplifying the geology.

At 30 cm resolution, only the thickest and most persistent units are recognized, with nearly all thin beds and sharp facies transitions eliminated. These differences are visually striking in the intervals noted above, where high-resolution data reveal geological heterogeneity that is completely unrecognized at coarse resolution. This focused comparison demonstrates that key geological features such as thin reservoir beds, baffles, or heterolithic zones are only detected with high-resolution sampling (1–3 cm). In intervals with complex stratigraphy, relying on coarser data can dramatically oversimplify the subsurface model, potentially leading to missed pay zones or misinformed reservoir management decisions.

Figure 8 shows the effect of Sampling Resolution on the Detection of Organic-Rich (TOC) Beds: Core and Lithofacies Comparison at 1 cm and 15 cm Resolution.

This figure compares core images with color-coded RockClass facies assignments for the same interval, using both high (1 cm, left) and conventional (15 cm, right) sampling resolutions. The 1 cm facies column accurately reproduces the complex alternation of thin beds and laminae seen in the core, particularly capturing numerous discrete black intervals that represent organic-rich (TOC) beds. These thin, dark features are critical for unconventional reservoir quality, as they often correspond to "sweet spots" with elevated total organic carbon, as also indicated by geochemical proxies such as Mo and U. (Zhou, 2012)

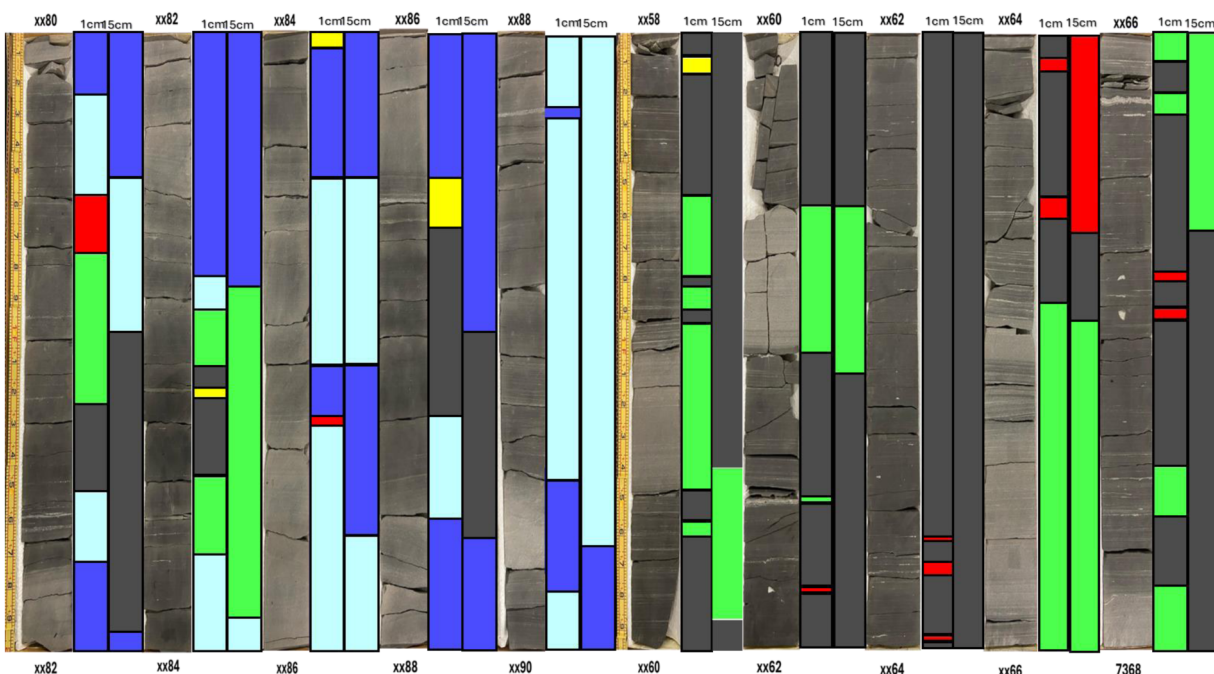


Figure 8—Facies logs at high resolution (1 cm, left) closely match thin beds and laminated features visible in the core, accurately highlighting organic-rich (black) and other thin facies. At lower resolution (15 cm, right), thin beds are merged or miss

At 15 cm resolution, however, thin TOC-rich beds and subtle facies changes are routinely merged with adjacent lithologies or lost entirely, resulting in a blocky, oversimplified representation of the true geological complexity. This underestimates both the abundance and continuity of organic-rich intervals, potentially leading to inaccurate reservoir evaluation and missed pay zones.

This comparison underscores that high-resolution sampling (1–3 cm) is essential for reliably identifying, mapping, and quantifying thin TOC-rich beds, ensuring that both geological heterogeneity and reservoir potential are faithfully captured.

Resolution, Acquisition Time, and Accuracy Trade-Offs

Figure 9 highlights the practical trade-off between sampling resolution, processing time, and classification accuracy in lithofacies analysis. At the finest resolution (1 cm), data collection and processing require significantly more time (over 130 hours in this study) but yield the highest classification accuracy (98%). By contrast, coarser resolutions such as 15 cm or 30 cm drastically reduce processing time but at the expense of accuracy, which drops to below 60% at 15 cm and just above 50% at 30 cm.

These results emphasize that while high-resolution sampling is resource-intensive, it is critical for achieving detailed and reliable geological interpretation. For many applications, 3 cm sampling offers a reasonable balance providing good accuracy (80%) with moderate processing demand while still preserving most key geological features. Ultimately, the choice of resolution should reflect both the study objectives and the acceptable trade-off between operational efficiency and interpretive confidence.

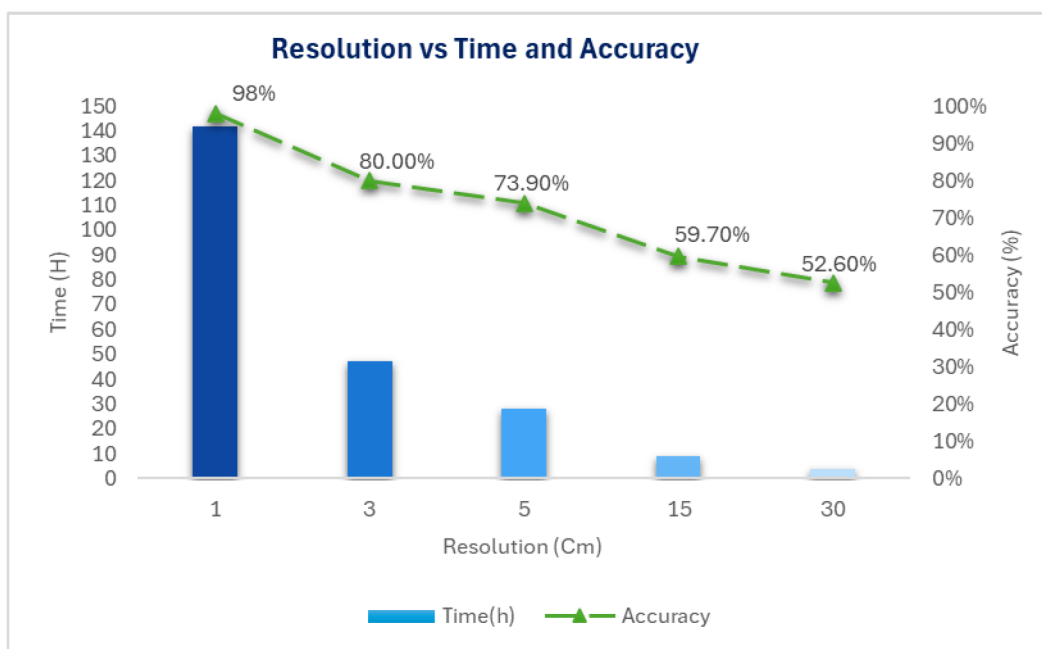


Figure 9—Effect of resolution on measurement time and accuracy. Higher resolution requires more time but yields better accuracy.

Conclusions

Fine-scale sampling at intervals of 3 cm or less is crucial for accurate facies classification, detection of TOC-rich intervals, and conducting advanced geochemical or petrophysical analyses in unconventional reservoirs. Such high-resolution data captures the thin beds and heterogeneities that are often key to understanding reservoir behavior and potential. In contrast, coarse resolutions are only appropriate for thick, homogeneous intervals or for preliminary screening, where detailed insight is not required. To support effective reservoir evaluation, sweet spot mapping, and completion design, high-resolution core scanning and XRF analysis should become standard practice. For advanced studies in unconventional reservoirs, especially those with significant thin bedding and geochemical variability, sampling at 1–3 cm intervals is strongly recommended. Coarser sampling should be approached with caution, as it increases the risk of facies misclassification and the loss of critical geological features. Additionally, all data processing workflows must preserve and document the sampling resolution, ensuring that interpretation methods are adjusted to reflect the level of detail available.

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